Detecting Drug Interactions with Machine Learning

Progress Report — v5

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Project Overview

Drug-drug interactions (DDIs) are a leading cause of adverse drug events and a critical challenge in pharmacovigilance, particularly in polypharmacy settings. This project explores machine learning-based DDI prediction, building on the DDI-LLM framework [1] which combines LLM embeddings with graph convolutional networks (GCNs) for link prediction on two benchmark datasets.

The central research question is: *which drug representation best captures interaction potential, and how do classical classifiers compare to GCN-based approaches?*

Related Work

Following the taxonomy of Qiu et al. [3], computational DDI detection methods fall into three categories: literature-based extraction (NLP on biomedical text), machine learning-based prediction (the focus of this project), and pharmacovigilance-based data mining (statistical signal detection from adverse event reports).

Within machine learning-based prediction, the most relevant prior works are:

DeepDDI [5] (Ryu et al., 2018) uses structural similarity profiles derived from drug molecular structures as input to a deep neural network. On DrugBank it reports AUC-ROC of 0.926 and AUPR of 0.919.

KGNN [4] (Lin et al., 2020) proposes a knowledge graph neural network that captures topological neighbourhood information from a biomedical KG. On DrugBank (V5.1.4, 2,578 drugs, 612,388 interactions) it achieves AUC-ROC of 0.991 and AUPR of 0.989 with the concat aggregation variant. On the OGB-biokg dataset it achieves ACC of 0.695 and AUC of 0.820.

LLM-DDI [2] (Li et al., 2026) is the most directly comparable work. It integrates LLM-generated molecular embeddings (text-embedding-ada-002, dimension 1536) into a GNN-based link prediction framework evaluated on two OGB benchmark datasets using 5-fold cross-validation. Key results from Table II are reproduced below:

Method	OGB-biokg ACC	OGB-biokg F1	OGB-biokg AUC	OGB-biokg AUPR	OGB-ddi ACC	OGB-ddi F1	OGB-ddi AUC	OGB-ddi AUPR
RF	0.910	0.911	0.972	0.973	0.730	0.733	0.804	0.787
KGNN	0.695	0.579	0.820	0.814	0.511	0.549	0.515	0.556
KG2ECapsule	0.937	0.938	0.982	0.971	0.735	0.752	0.794	0.744
LLM-DDI	**0.964**	**0.964**	**0.994**	**0.994**	**0.822**	**0.824**	**0.900**	**0.883**

The results of this project will be directly compared against this table. Note that LLM-DDI uses OGB-biokg and OGB-ddi as benchmarks, which motivates using these same datasets in the current project (Supervisor feedback, point 2).

Datasets

Current datasets (DDI-LLM repo)

Dataset	Drugs	Interactions	Sparsity	After filtering
BioSnap	1,514	48,514	4.2%	~27,800 edges
DrugBank	1,706	191,808	13.2%	—

Important finding: the DDI-LLM notebook filters out drug pairs where either drug lacks a valid SMILES string *or* a valid description, reducing BioSnap from 48,514 to ~27,800 edges. This filter is unnecessarily applied even for description-only embeddings. Removing this constraint for text-based embeddings is a methodological improvement implemented in this project.

Extended datasets (to be added per supervisor feedback)

OGB-biokg — a heterogeneous biomedical knowledge graph from the Open Graph Benchmark containing 81,422 entities (drugs, proteins, diseases, side effects) and 3,155,786 triplets including 139,345 DDIs across 9,078 unique drugs.

OGB-ddi — a homogeneous DDI graph of 4,267 FDA-approved or experimental drugs with 1,067,911 triplets across 20 relation types. Uses a protein-target split ensuring the test set contains drugs targeting different proteins than training.

Vivi spoke about OGB-ddi in March.

Negative samples — CRESCENDDI

The original DDI-LLM pipeline generates negative samples by randomly sampling drug pairs absent from the interaction list. This is methodologically weak: an unlabelled pair may be an undiscovered positive interaction rather than a true negative.

CRESCENDDI [8] is a publicly available reference set containing 10,286 positive and **4,544 confirmed negative controls** covering 454 drugs and 179 adverse events, available on Figshare. The negative controls were identified by scanning the scientific literature to confirm the *absence* of evidence for an interaction, making them clinically verified true negatives.

Mapping to DrugBank IDs

CRESCENDDI identifies drugs by normalised RxNorm names (e.g. *valproate*), whereas BioSnap and DrugBank use DrugBank identifiers (e.g. `DB00313`). A mapping was performed in two steps. First, a lookup dictionary was built from the DrugBank vocabulary file (`drugbank_vocabulary.csv`, 19,853 drugs) by indexing both the primary common name and all listed synonyms, yielding

55,576 name–ID entries. Second, each drug name in CRESCENDDI was normalised (lowercase, whitespace stripped) and looked up in this dictionary.

Results: 4,535 out of 4,544 negative pairs (99.8%) were successfully mapped. The 9 unmapped pairs all correspond to the truncated name *mefenamate* (correct form: *mefenamic acid*), which is absent from the DrugBank synonym list and was discarded.

Coverage on working datasets

	BioSnap	**DrugBank**
Unique drugs in CRESCENDDI	434	
CRESCENDDI drugs found in dataset	414 / 434 (95.2%)	412 / 434 (94.7%)
Negative pairs usable	**4,306 / 4,535 (95.0%)**	**4,078 / 4,535 (89.9%)**

The 229 unusable pairs in BioSnap correspond to the 20 CRESCENDDI drugs absent from the BioSnap graph. These pairs are discarded since the GCN cannot produce embeddings for unseen nodes.

Class imbalance strategy

The 4,306 confirmed negatives available for BioSnap are substantially fewer than the ∼27,800 positive interactions (ratio ≈ 1:6). The strategy adopted combines CRESCENDDI negatives with complementary random sampling to restore balance, using a weighted BCE loss to correct residual imbalance:

```
pos_weight = torch.tensor([n_neg / n_pos])  # ~6.5 for BioSnap
criterion = torch.nn.BCEWithLogitsLoss(pos_weight=pos_weight)
```

This represents a methodological improvement over the original pipeline, which relied entirely on random negative sampling.

Data Analysis

Coverage summary

	BioSnap		**DrugBank**	
Feature	Count	%	Count	%
Valid descriptions	1,489/1,514	98.3%	1,695/1,706	99.4%
Corrupted descriptions	8/1,514	0.5%	6/1,706	0.4%
Missing descriptions	17/1,514	1.1%	5/1,706	0.3%
Valid SMILES	1,332/1,514	88.0%	1,695/1,706	99.4%

Two corruption types found in the description file: `;;;;` (n = 6,512) and `#NAME?` (n = 6), both Excel artefacts. Only 14 drugs across both datasets are affected. Zero-vector fallback strategy applied.

Description length

Median description length is 90 characters with high variance (std = 245). The wide range suggests that drugs with very short descriptions will produce weaker LLM embeddings, a limitation to be discussed in the final report.

Methodology

Pipeline

Each drug is described by a SMILES string (chemical structure) and a text description (clinical information). These are fed into an embedding model to produce a feature vector per drug. Feature vectors are attached as node features to the DDI interaction graph. A 3-layer GCN encoder produces latent drug representations, and a dot-product decoder predicts interaction probability for any drug pair.

Embeddings compared

Embedding	Input	Type	Status
Morgan Fingerprints	SMILES	Classical structural	To generate
BERT	Text description	General LLM	Pre-computed
GPT	Text description	General LLM	Pre-computed
LLaMA / LLaMA2	Text description	General LLM	Pre-computed
PubMedBERT	Text description	**Biomedical LLM**	**To generate**
ChemBERTa	SMILES	**Chemistry LLM**	**To generate**

Comparison approaches

Classical baselines — drug pair embeddings fed into logistic regression, random forest, and MLP (scikit-learn). These require no graph structure and serve as the conventional ML comparison.

GCN-based link prediction — the DDI-LLM pipeline extended with new embeddings, CRESCENDDI negatives, and the OGB datasets.

Overfitting analysis (supervisor feedback, point 3)

In addition to reporting test metrics, training vs. validation loss curves will be plotted for each configuration to detect overfitting. An error analysis will identify which drug pairs the model consistently misclassifies and whether these correspond to drugs with low degree in the graph or short/poor descriptions.

Statistical rigour

Each configuration: minimum 5 random seeds, results reported as mean $\pm$ std. Significance between embedding types: paired Wilcoxon test. Primary metrics: ACC, F1, AUROC, AUPR (to match LLM-DDI Table II).

Environment

Model training and embedding generation are performed on Google Colab Pro (GPU). To avoid reinstalling packages at every session, all dependencies are installed once to a persistent folder on Google Drive and added to the Python path at session start:

```
import sys
sys.path.insert(0, "/content/drive/MyDrive/colab_packages")
```

Local development (data exploration, script editing) uses VS Code on Windows 11, Python 3.12, with a `.venv` virtual environment. Key packages: `torch==2.5.1`, `torch_geometric==2.7.0`, `transformers==5.3.0`, `numpy==1.26.4`, `pandas==3.0.1`, `rdkit`, `scikit-learn`, `ogb`.

Supervisor Feedback and Response

Feedback	Response / Action
Add CRESCENDDI for confirmed negative interactions	Download from Figshare; replace random negative sampling; use weighted BCE loss for class imbalance
Combine ogbl-ddi + DrugBank for more drugs	Load via OGB Python package; enables direct comparison with LLM-DDI Table II (p.778)
GCN overfitting analysis	Add train/val loss curves; error analysis on misclassified pairs; identify low-degree and short-description drugs
Compare with January 2026 paper (p.778)	Target: LLM-DDI on OGB-biokg (AUC 0.994) and OGB-ddi (AUC 0.900); same metrics (ACC, F1, AUC, AUPR), same 5-fold cross-validation

Action Plan for Next Meeting

Week	Task	Details
1	*CRESCENDDI*	Downloaded, mapped (99.8%), 4,306 usable negatives for BioSnap, 4,078 for DrugBank. **Done.**
1	Load OGB datasets	`pip install ogb`; load ogbl-ddi and ogbl-biokg; inspect structure and compare with current datasets
1–2	Reproduce DDI-LLM baseline	Run existing notebook on BioSnap with precomputed BERT/GPT embeddings; verify results match published numbers
2	Implement classical baselines	Logistic regression, random forest, MLP on concatenated drug pair embeddings (scikit-learn)
2	Add overfitting diagnostics	Plot train/val loss curves; identify misclassified drug pairs; check correlation with description length and node degree
3–4	Generate new embeddings	Morgan fingerprints (RDKit); PubMedBERT (`microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract`); ChemBERTa (`seyonec/ChemBERTa-zinc-base-v1`)
Next meeting		Present: (1) CRESCENDDI integration status, (2) OGB dataset loaded and explored, (3) baseline results on BioSnap, (4) first overfitting curves

Open question for next meeting

For the comparison with LLM-DDI (Table II, p.778), should we follow exactly the same evaluation protocol (5-fold cross-validation), or can we adapt the protocol to fit our pipeline?

References

[1] S. Shaghayegh et al., *DDI-LLM: Drug-Drug Interaction Prediction via Large Language-Based Drug Information Embedding for Multi-View Representation Learning*, https://github.com/sshaghayeghs/DDI-LLM, 2023.

[2] X. Li et al., *LLM-DDI: Leveraging Large Language Models for Drug-Drug Interaction Prediction on Biomedical Knowledge Graph*, IEEE Journal of Biomedical and Health Informatics, Vol. 30, No. 1, pp. 778–789, 2026.

[3] Y. Qiu et al., *A Comprehensive Review of Computational Methods for Drug-Drug Interaction Detection*, IEEE/ACM Transactions on Computational Biology and Bioinformatics, 19(4), 2022.

[4] X. Lin et al., *KGNN: Knowledge Graph Neural Network for Drug-Drug Interaction Prediction*, IJCAI, 2020.

[5] J. Y. Ryu, H. U. Kim, and S. Y. Lee, *Deep Learning Improves Prediction of Drug-Drug and Drug-Food Interactions*, PNAS, 115(18), 2018.

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[8] E. Vilar et al., *A Reference Set of Clinically Relevant Adverse Drug-Drug Interactions*, Scientific Data, 2022. https://doi.org/10.1038/s41597-022-01159-y